

OM PRATAP TILWAR

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Education

Indian Institute of Information Technology, Nagpur

August 2023 - May 2027

B.Tech in Electronics and Communication Engineering (IoT) – **CGPA: 8.36**

Relevant Coursework: Digital Signal Processing, Linear Algebra, Probability & Statistics, Data Structures & Algorithms.

Technologies and Skills

Medical Imaging & CV: Image Classification, Segmentation (U-Net), Object Detection (YOLOv8), DICOM, CLAHE, Grad-CAM.

ML/DL Frameworks: PyTorch, OpenCV, Scikit-Learn, TensorFlow, Transformers, ONNX Runtime.

Programming: Python, C, C++, SQL, MATLAB.

Tools & Deployment: FastAPI, Docker, Git, Linux, Data Preprocessing, Feature Engineering.

Domain: Healthcare Data Privacy (HIPAA), PHI De-identification, DICOM PS3.15, OCR, Biomedical NLP.

Experience

Research Intern, IIIT – Nagpur, Maharashtra

August 2025 – February 2026

- Developed deep learning models (YOLOv8) for dense object detection, performing **data preprocessing** and feature engineering on 3,200+ annotated instances to improve mAP by **8.4%** – transferable to medical lesion detection.
- Built end-to-end ML pipelines from data ingestion to model training, conducted systematic experiments across hyperparameter configurations, and documented results for **reproducibility**.
- Designed inference pipelines with ONNX export and **INT8 quantization** for edge deployment, achieving **<30ms** latency – applicable to **point-of-care** medical imaging devices.
- Published findings in the **IEEE Xplore** digital library (April 2026): [IEEE Document 11448187](#).

Projects

MedVision: Medical Imaging AI Toolkit | Python, PyTorch, OpenCV, FastAPI, Docker



- Fine-tuned ResNet-18 classifier on **5,800+ real chest X-rays** with differential learning rates and layer freezing, achieving **91.8% accuracy**, **0.97 AUC-ROC**, and **92.3% sensitivity** on a 624-image held-out test set.
- Built U-Net segmenter from scratch (17M params, Dice+BCE loss, cosine annealing) achieving **0.83 Dice** / 0.71 IoU; implemented medical preprocessing with **CLAHE** enhancement and CT windowing presets (lung, bone, brain).
- Developed **HIPAA-compliant** DICOM de-identification following **PS3.15 Annex E** (25+ PHI tags, UID regeneration) and OCR pipeline with regex-based **NER** for patient name, MRN, date, and SSN detection.
- Created **Grad-CAM** interpretability for per-class attention maps; deployed as Docker-containerized **FastAPI** REST API (6 endpoints), validated with **48 automated tests**.

AuthNet: Deep Metric Learning for Product Authentication | Python, PyTorch, ONNX



- Trained a ResNet-18 backbone with **Multi-Similarity Loss** and hard-negative mining to learn 128-dim embeddings across 47 texture classes, achieving **64.5% Recall@1** (31x above random), 48.97% mAP, and 0.70 NMI on 1,880 test samples.
- Implemented custom **Grad-CAM** for embedding architectures and **t-SNE** visualization confirming clear cluster separation; built cosine-similarity authentication API with confidence scoring.
- Optimized for edge deployment via **ONNX Runtime** with INT8 post-training quantization (**4x compression**, 43.3 MB → 10.9 MB, lossless: max diff < 8e-8), achieving **2.4ms** GPU / 8ms CPU inference.

FinSight AI: LLM-Powered RAG Pipeline | Python, FastAPI, LLM, Pinecone



- Engineered an **LLM-based** conversational AI system integrating a local Llama-3 model with **prompt engineering** and context injection for privacy-first inference — no data leaves the device.
- Built a 4-tier hybrid **RAG pipeline**: TF-IDF keyword matching → Pinecone **vector search** → deterministic function-calling → LLM generation, achieving **sub-second** retrieval across structured banking data.

Achievements

- Winner** – Bioinformatics Machine Learning Challenge, IIIT Nagpur 2025 (300+ participants).
- Winner** – Sankalp Bharat Hackathon (Machine Learning), 2026 (976 participants).
- Winner** – Machine Learning Hackathon, IIIT Nagpur, 2026 (850+ participants).
- 7th among 800+ teams in eYantra Robotics Competition (eYRC), IIT Bombay (2024-25).